Symmetry-Aware Reinforcement Learning for DDoS and EDoS Attack Detection and Mitigation:

A MatrixNet Approach to Algebraic Traffic Fingerprinting

Kevin Lee

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Abstract

Modern Distributed Denial of Service (DDoS) and Economic Denial of Sustainability (EDoS) attacks increasingly rely on *obfuscation*: varying their surface-level traffic characteristics (source IPs, packet sizes, inter-arrival times) to evade signature- and threshold-based detectors. This proposal develops a fundamentally different defense paradigm, drawing on the recently developed *MatrixNet* framework [1], which learns symmetry-invariant neural representations of sequential algebraic structures. The central idea is that network traffic sequences can be modeled as *words in a formal algebraic structure*, and that attacks of the same class—no matter how obfuscated—share an invariant algebraic fingerprint analogous to the *Garside normal form* in braid group theory [2]. A reinforcement learning agent, equipped with these symmetry-aware representations, can then learn adaptive mitigation policies that generalize to unseen attack variants. The relevant literature is surveyed, the novelty of the proposed approach is established, the key research questions are identified, and a research and evaluation plan is outlined.

Contents

1 Introduction and Motivation **3**
1.1 The Core Problem: Obfuscation Defeats Current Defenses . . . 3
1.2 The Key Insight: Attacks Have Algebraic Invariants . . . 3
1.3 Scope of This Proposal . . . 3

2 Background and Related Work **4**
2.1 DDoS Detection: From Rules to Deep Learning . . . 4
2.2 EDoS Attacks: Stealthy, Economic, Underexplored . . . 4
2.3 Reinforcement Learning for Adaptive Mitigation . . . 4
2.4 Symmetry-Invariant Neural Networks . . . 5
2.5 Algebraic Structures in Security . . . 5

3 Proposed Framework **6**
3.1 Overview . . . 6
3.2 Defining the Algebraic Structure of Traffic . . . 6
3.3 MatrixNet as Attack Encoder . . . 6
3.4 Attack Invariants as the Canonical Form . . . 7
3.5 The RL Mitigation Agent . . . 7
3.6 Framework Diagram . . . 9

4 Research Questions **9**

5 Expected Contributions **10**

6 Related Work Summary Table **10**

7 Datasets and Evaluation Plan **11**
7.1 Datasets . . . 11
7.2 Evaluation Metrics . . . 11
7.3 Baselines . . . 11

8 Research Timeline **12**

9 Risks and Mitigations **12**

10 Connection to Companion Paper **13**

11 Conclusion **13**

1 Introduction and Motivation

1.1 The Core Problem: Obfuscation Defeats Current Defenses

A Distributed Denial of Service (DDoS) attack floods a target with traffic to exhaust computational or network resources. Its cousin, Economic Denial of Sustainability (EDoS) [5], targets cloud-hosted services by triggering expensive auto-scaling operations through carefully crafted, low-volume traffic—draining the victim's budget rather than bandwidth.

The state-of-the-art defense pipeline—volume thresholds, deep-packet inspection, machine learning classifiers trained on known signatures—shares a fundamental weakness: it detects the *surface form* of an attack. A sophisticated attacker who randomizes source IP addresses, varies packet sizes, and distributes the attack across thousands of bots can often evade all of these detectors simultaneously [16]. Adversarial machine learning studies show that even strong deep-learning classifiers drop from over 99% accuracy to under 50% accuracy under adaptive obfuscation [16, 17].

1.2 The Key Insight: Attacks Have Algebraic Invariants

Despite their surface variation, attacks of the same class share *deep structural properties*: they ultimately target the same resource (bandwidth, compute, money), follow the same protocol exploitation pattern, and—crucially—generate traffic sequences whose *algebraic composition* is invariant across disguises.

An analogy to the *braid group* in mathematics [3] is instructive. In braid theory, infinitely many different *words* (sequences of generators) can describe the same underlying braid element. The *Garside normal form* is a canonical representation that cuts through this surface ambiguity: two words are equivalent if and only if they reduce to the same Garside form [2]. Recent work on *MatrixNet* [1] shows that neural networks can *learn* representations that respect this algebraic invariance, even without being given the group structure in advance.

The same principle is proposed to apply to attack traffic: there exists a *canonical attack fingerprint*—an invariant algebraic structure—that persists across obfuscation variants, and that MatrixNet-style learning can discover it.

1.3 Scope of This Proposal

This proposal focuses on two attack classes where the gap between current defenses and practical security is most acute:

(1) **Volumetric and low-rate DDoS attacks** in Software-Defined Networking (SDN) environments, where RL-based mitigation (rate limiting, rerouting, blocking) is the natural response framework [11, 12].

(2) **EDoS attacks on cloud APIs**, where individual requests look legitimate but their sequential composition triggers runaway auto-scaling costs [5, 6].

2 Background and Related Work

2.1 DDoS Detection: From Rules to Deep Learning

Traditional rule-based defenses (rate limiting, IP blacklisting) fail against distributed, low-rate, or obfuscated attacks. The field has moved to machine learning: early work used SVMs and random forests on NetFlow features [20], then convolutional and recurrent networks for temporal pattern capture [22, 21].

The current state of the art applies *Graph Neural Networks (GNNs)* to traffic: endpoints become nodes, flows become edges, and GNNs learn relational patterns invisible to per-packet classifiers [9, 10]. Reported accuracy exceeds 99% on standard benchmarks. However, these models are brittle: targeted adversarial perturbations (TCP segmentation, packet reordering, timing jitter) dramatically degrade performance [16, 17].

2.2 EDoS Attacks: Stealthy, Economic, Underexplored

EDoS was formalized by Sqalli et al. [5]: an attacker submits workloads that appear legitimate but are designed to force expensive cloud operations (VM instantiation, database queries, transcoding jobs). The economic damage accumulates over days or weeks, far below any single-threshold alarm.

Detection work has used multihead attention networks (MAN-EDoS [6]), LSTM + GAN anomaly detection [8], and random forests [7], achieving 92–98% accuracy. Critically, all of these approaches rely on *volume or rate features*—they detect that something unusual is happening, not *what type of attack* is unfolding. They are therefore susceptible to attackers who stay below every rate threshold on every feature.

2.3 Reinforcement Learning for Adaptive Mitigation

RL has emerged as a compelling mitigation paradigm because mitigation is inherently sequential: the defender applies one action at a time (block IP, rate limit flow, reroute traffic)

and must balance false positives against responsiveness. Key systems include DAMS [11] (Double DQN for SDN flow-table modification), QoSentry [12] (QoS-preserving mitigation via RL), and Q-MIND [13] (stealthy DoS defense in SDN). All of these systems define state as raw NetFlow features and action as SDN control commands.

A persistent limitation is *poor generalization to unseen attack variants*: because state is defined as surface-form features, a policy trained on known attacks fails when the attacker changes strategy [16].

2.4 Symmetry-Invariant Neural Networks

Group-equivariant convolutional networks (G-CNNs), introduced by Cohen and Welling [14], build symmetry invariance directly into the network architecture. When the symmetry group is known (e.g., rotations, reflections), G-CNNs dramatically reduce sample complexity. Applications in security include website fingerprinting with equivariant/invariant feature learning [15].

MatrixNet [1] generalizes this idea to settings where the symmetry group is *unknown*: it learns a matrix representation $\rho : G \to \mathrm{GL}(d, \mathbb{R})$ end-to-end, supervised only by a relation loss $\mathcal{L}_{\mathrm{rel}} = \sum_{r_i \in R} \|M(r_i) - I\|_F^2$ that enforces the group axioms. Validated on the symmetric group and the Artin braid group, MatrixNet achieves higher sample efficiency and stronger generalization than baselines. No prior work applies MatrixNet—or any learned group representation—to network security.

2.5 Algebraic Structures in Security

Linear algebra has been applied to network intrusion detection (covariance matrices of flow features, spectral analysis of adjacency matrices [18]). Garside normal forms and braid groups appear in post-quantum cryptography [19] but not in traffic analysis. To the best of the author's knowledge, no prior work models network traffic as a word in a formal algebraic structure with computable canonical forms.

Novelty Gap

The gap: No existing system combines (1) learned symmetry-invariant traffic representations, (2) algebraic fingerprinting of attack sequences, and (3) RL-based adaptive mitigation. Each component has strong individual precedent; the integration is entirely novel.

3 Proposed Framework

3.1 Overview

The proposed system has three tightly coupled layers, directly mirroring the braid-group framework of the companion paper [4]:

Layer 1. Algebraic Traffic Encoding (MatrixNet layer). Model a time-windowed traffic sequence as a *word* $w = e_{t_1} e_{t_2} \cdots e_{t_\ell}$ in a vocabulary of *traffic event types* (generators). Assign each generator a learned invertible matrix; compose them sequentially to obtain a symmetry-aware state embedding $s(w) = \mathrm{flat}(M(w))$, where $M(w) = \prod M_{e_{t_k}}$.

Layer 2. Algebraic Fingerprint Extraction (invariant layer). Train MatrixNet to predict *attack invariants*—scalar quantities that are the same for all surface-level variants of the same attack class (analogous to Jordan–Hölder multiplicities in the braid setting). These invariants serve as the attack "canonical form" and can be compared against a library of known attack fingerprints.

Layer 3. Reinforcement Learning Mitigation (policy layer). An RL agent receives $s(w)$ as its state and selects mitigation actions. Because $s(w)$ is approximately invariant to obfuscation of the same attack class, the policy generalizes across disguise variants without retraining.

3.2 Defining the Algebraic Structure of Traffic

Let $\mathcal{G} = \{g_1, \ldots, g_K\}$ be a finite vocabulary of *traffic event generators*, each representing a (discretized) combination of protocol, direction, and size bin. A traffic window of length ℓ is a word $w = g_{i_1} g_{i_2} \cdots g_{i_\ell} \in \mathcal{G}^*$. It is posited that traffic events compose according to an approximate algebraic structure: certain pairs of events commute (e.g., independent flows from different sources), and certain triples satisfy approximate braid-like relations reflecting protocol handshake patterns. These relations are learned by MatrixNet from labeled data.

3.3 MatrixNet as Attack Encoder

For each generator g_k, MatrixNet learns a matrix $M_{g_k} \in \mathrm{GL}(d, \mathbb{R})$ via the matrix-exponential construction:

$$M_{g_k} = \exp(A_k), \quad A_k = \mathrm{Linear}(v_{g_k}), \tag{1}$$

where v_{g_k} is a signed one-hot encoding of generator g_k. Invertibility is guaranteed by the matrix exponential. The representation of a traffic word is:

$$M(w) = M_{g_{i_1}} \cdot M_{g_{i_2}} \cdots M_{g_{i_\ell}}. \tag{2}$$

Two traffic sequences that are obfuscated variants of the same attack class are expected to produce (approximately) the same $M(w)$. The relation loss

$$\mathcal{L}_{\text{rel}} = \sum_{r_j \in \mathcal{R}} \|M(r_j) - I\|_F^2 \tag{3}$$

penalizes violations of learned traffic relations $\mathcal{R}$, enforcing algebraic consistency throughout training.

3.4 Attack Invariants as the Canonical Form

Analogous to Jordan–Hölder multiplicities (algebraic invariants of braid elements predicted by MatrixNet [1]), *attack invariants* are defined as scalar quantities extracted from $M(w)$ that are (approximately) constant across obfuscation variants of the same attack class. Candidates include:

- The *spectral signature*: sorted eigenvalue magnitudes of $M(w)$.
- The *trace vector*: $[\text{tr}(M(w)), \text{tr}(M(w)^2), \text{tr}(M(w)^3)]$ (analogous to power-sum symmetric functions, invariant to conjugation).
- The *Frobenius distance to identity*: $\|M(w) - I\|_F$, measuring how far the traffic pattern is from "normal" (for which $M(w) \approx I$).

The canonical normal form is defined implicitly: a traffic sequence is "benign" if its invariants lie within a learned benign region, and "attack of class k" if its invariants match the fingerprint of class k.

3.5 The RL Mitigation Agent

State. $s_t = s(w_t) = \text{flat}(M(w_t)) \in \mathbb{R}^{d^2}$, the MatrixNet embedding of the current traffic window.

Action space.

(a) **Do nothing** (monitor).

(b) **Rate-limit** a source IP class.

(c) **Block** an IP range (CIDR block).

(d) **Reroute** to a scrubbing center.

(e) **Issue CAPTCHA / challenge** to suspicious sessions.

(f) **Throttle** a specific API call class (for EDoS).

(g) **Trigger alert** and escalate to human operator.

Reward function.

$$r_t = r_{\text{det}} + \alpha\, r_{\text{fp}} + \beta\, r_{\text{lat}} - \delta, \tag{4}$$

where $r_{\text{det}} = +1$ for each confirmed attack flow mitigated, $r_{\text{fp}} = -5$ for each falsely blocked legitimate request (high penalty to protect user experience), r_{lat} rewards low-latency decisions, and $-\delta = -0.05$ is a per-step cost to discourage inaction. A terminal reward of $+10$ is given when the network returns to a confirmed benign steady state.

Key property. Because $s(w)$ is approximately invariant to obfuscation, the policy $\pi_\theta(a \mid s)$ trained on known attack variants will assign similar probabilities to the same mitigating action for unseen variants of the same class—without any retraining.

3.6 Framework Diagram

4 Research Questions

Research Questions

RQ1. Algebraic regularity: Do real-world DDoS and EDoS traffic sequences exhibit approximate algebraic regularities (commutativity, braid-like relations) that are stable across obfuscation variants of the same attack class?

RQ2. Representation learning: Can MatrixNet learn symmetry-invariant representations of traffic sequences that cluster same-class attacks regardless of surface obfuscation, and separate them from benign traffic?

RQ3. Generalization: Does an RL policy trained on known attack variants, using MatrixNet state embeddings, generalize to *unseen* variants—including adaptively obfuscated attacks—more effectively than policies trained on raw NetFlow features or GNN embeddings?

RQ4. EDoS fingerprinting: Do the spectral invariants of $M(w)$ serve as reliable canonical fingerprints for EDoS attack classes, enabling detection below all individual rate thresholds?

RQ5. Computational feasibility: Can the MatrixNet encoding pipeline operate in real time ($\leq 10\,\text{ms}$ per window) on commodity hardware, or does the matrix product construction require approximation for practical deployment?

5 Expected Contributions

Expected Contributions

C1. Algebraic traffic model: A formal definition of network traffic sequences as words in a generator alphabet, with a learned approximate group structure. This is the first application of formal algebraic sequence modeling to network security.

C2. MatrixNet traffic encoder: An adaptation of the MatrixNet architecture [1] to the traffic domain, including a new generator vocabulary design, relation loss calibration for approximate (statistical) traffic relations, and attack invariant extraction from the learned matrices.

C3. Symmetry-aware RL defense agent: An actor-critic RL agent that uses MatrixNet embeddings as state, with a principled reward function balancing detection rate, false positive cost, and latency. Evaluated on DDoS and EDoS datasets.

C4. Generalization benchmark: A new evaluation protocol testing detection accuracy under adaptive obfuscation, comparing MatrixNet-RL against GNN-RL [10] and attention-based baselines [6].

C5. Open-source release: Code, trained models, and the evaluation benchmark will be released to the community.

6 Related Work Summary Table

Table 1 positions the proposed work against the most closely related prior systems along five key dimensions.

Table 1: Comparison of related work. ✓= feature present, ≈ = partial, × = absent.

System	Sym.-Inv.	Alg. FP	RL Policy	EDoS	Obfusc. Robust
GraphDDoS [9]	×	×	×	×	×
GraphFedAI [10]	≈	×	×	×	×
DAMS [11]	×	×	✓	×	×
QoSentry [12]	×	×	✓	×	×
MAN-EDoS [6]	×	×	×	✓	×
MatrixNet (math) [1]	✓	✓	×	×	n/a
Proposed	✓	✓	✓	✓	✓

Sym.-Inv. = symmetry-invariant representation; *Alg. FP* = algebraic fingerprinting; *RL*

Policy = RL-based mitigation; *EDoS* = EDoS-capable; *Obfusc. Robust* = validated under obfuscation.

7 Datasets and Evaluation Plan

7.1 Datasets

- **CIC-DDoS2019** [23]: 87 GB of labeled DDoS traffic across 12 attack types (SYN flood, UDP flood, HTTP flood, slowloris, etc.) from the Canadian Institute for Cybersecurity. Primary benchmark for DDoS detection.
- **CIC-IDS2017** [24]: Multi-day capture including brute-force, web attacks, infiltration, and DoS. Used for evaluating generalization to unseen attack types.
- **CAIDA DDoS 2007** [25]: Backbone-level DDoS trace; used to test scalability of encoding pipeline.
- **Synthetic EDoS traces**: Controlled EDoS scenarios will be generated against a cloud microservice testbed (AWS Lambda + API Gateway), varying the injection rate, request complexity, and disguise strategy.
- **Obfuscation benchmark**: Following Apruzzese et al. [16], six obfuscation transforms (IP randomization, timing jitter, payload padding, protocol tunneling, fragmentation, flow interleaving) will be applied to the above datasets to evaluate performance degradation.

7.2 Evaluation Metrics

- **Detection:** Accuracy, precision, recall, F1-score, area under ROC curve (AUC-ROC).
- **Obfuscation robustness:** Accuracy at each of the six obfuscation levels vs. baseline systems.
- **RL mitigation:** Mean episode reward, time to benign steady state, false-positive rate, throughput impact.
- **Efficiency:** Per-window encoding latency (ms), GPU memory, throughput (windows/second).

7.3 Baselines

(1) GNN classifier (GraphFedAI [10]) + threshold-based mitigation.

(2) GNN classifier + DQN mitigation (DAMS [11]).

(3) MAN-EDoS attention network [6] for EDoS.

(4) Raw-feature DQN (no symmetry-aware encoding).

(5) MatrixNet encoder + threshold (no RL).

The proposed full system (MatrixNet + RL) is expected to outperform all baselines on obfuscation robustness (RQ3) while remaining competitive on accuracy for non-obfuscated attacks.

8 Research Timeline

Table 2: Proposed research timeline (12-month plan).

Month(s)	Milestone	Deliverable
1–2	Literature review; finalize algebraic traffic model	Report
2–3	Generator vocabulary design; dataset preprocessing	Code + docs
3–5	MatrixNet adaptation + training on DDoS data	Trained model
5–6	Attack invariant extraction; fingerprint validation (RQ1, RQ2)	Paper section
6–8	RL environment implementation; reward tuning	RL agent
8–9	EDoS testbed setup; synthetic trace generation (RQ4)	Dataset
9–10	Obfuscation benchmark evaluation (RQ3, RQ5)	Results
10–11	Comparison against all baselines; ablation study	Results table
11–12	Paper writing; code release; submission	Full paper

9 Risks and Mitigations

R1. Traffic may not exhibit algebraic regularity (RQ1). *Mitigation:* If exact group relations do not emerge, the framework falls back to a *monoid* structure (no inverses required), which is sufficient for the MatrixNet encoding and still provides symmetry-invariant representations.

R2. MatrixNet may be too slow for real-time use (RQ5). *Mitigation:* Approximate matrix products (e.g., fixed-rank approximation, block-diagonal structure as in MatrixNet-MC [1]) and batch-parallel GPU encoding will be explored.

R3. RL training instability. *Mitigation:* Proximal Policy Optimization (PPO) [26] with careful reward normalization will be used. If RL proves unstable, supervised

imitation learning from expert traces provides a fallback policy.

R4. EDoS synthetic data may not generalize to real deployments. *Mitigation:* Validation on real cloud billing data (if available under NDA) and sensitivity analysis with respect to trace generation parameters will be conducted.

10 Connection to Companion Paper

This proposal is a direct extension of the companion paper [4], which develops the theoretical framework of RL + MatrixNet + Garside normalization for the mathematical problem of braid group word reduction. That paper establishes:

- The MatrixNet architecture and its relation loss for enforcing algebraic consistency.
- The RL reward design for sequential canonical-form navigation.
- The role of Jordan–Hölder multiplicities as algebraic invariants that MatrixNet learns to predict.

The present proposal *translates* each of these elements to the network security domain: traffic windows replace braid words, attack invariants replace Jordan–Hölder multiplicities, benign steady state replaces Garside normal form, and SDN control actions replace generator moves. The mathematical scaffolding is the same; the application domain is new.

11 Conclusion

This proposal has outlined a research agenda for a novel DDoS/EDoS defense framework grounded in the algebraic structure of network traffic. The core hypothesis—that attacks of the same class share invariant algebraic fingerprints discoverable by MatrixNet-style learning, and that RL agents equipped with these representations generalize to obfuscated variants—is both theoretically motivated and practically testable. No prior work integrates symmetry-invariant representation learning, algebraic fingerprinting, and RL-based adaptive mitigation. This work represents a genuine contribution to both the network security and geometric machine learning communities, with potential publication at USENIX Security, ACM CCS, or NeurIPS.

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